An IoT-Based Smart Doorbell System Using Camera and Ultrasonic Sensor for Real-Time Visitor Detection and Context-Aware Monitoring

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ABSTRACT-

The Smart Doorbell System is designed to enhance home security by combining embedded Python control with Arduino-based sensor management. The setup uses an ultrasonic sensor to detect visitor presence and a USB camera to capture real-time images, which are immediately sent to the homeowner through Telegram for remote verification. After reviewing the image, the user can grant access by sending a confirmation message, prompting the system to activate the voice module and operate the motor driver to unlock and open the door. The door remains open for a preset time of 20 seconds before automatically closing. Additional safety elements such as a GSM module, buzzer, and red LED provide alert notifications during unauthorized access or emergency events. Powered through a stable 12V adapter and interconnected using multiple connectors, the smart doorbell offers a reliable, interactive, and secure solution for modern home entry management.

***Keywords*:** Smart Doorbell System, Embedded Python Control, Arduino-Based Security, Telegram Remote Access, IoT Home Automation.

I. INTRODUCTION

The suggested system brings a smart home security system which entails the application of the anomaly detection process as well as face recognition process within the framework of the IoT-enabled system. The model is based on the logit-boosted Convolutional Neural Networks (CNNs) to enhance the precision of the classification and the detection in the real-time applications. The system recognizes deviant behavior and intrusion into the system through continuous inspections of video feeds and device usage trends. The face recognition system identifies those who are using the space that is being monitored and the anomaly detection system is able to monitor the alterations of the common working plans of the smart home appliances. The boosting process of the logit increases the learning strength of the CNN, through refining the misclassification cases, which results in the robustness and reliability. The system is designed to be executed on devices with limited resources of the IoT setting that ensures the effective processing of information, the security of the data, and the rapid reaction to the threat of security concerns. The provided solution presents a scalable, automated, and flexible solution to the existing smart home environments, and this enhances the security and ease to the user [1].

The IoT-Based Smart Home Security Alert System to accomplish the Continuous Supervision is designed to ensure the provision of the real-time monitoring and

the immediate notification of the threat in the residential environment. The system integrates sensors that are linked to one another to offer multiple monitoring services of the home premises in real-time through the communication and cloud-based services. The parameters of the environment and security (motion, door, gas leakage, and intrusion) are continuously monitored with the help of the Internet of Things (IoT) technology. The system automatically alerts the owner of the house whenever an abnormal activity is detected by sending him or her warning systems such as mobile messages, alarm clocks, or SMS messages. A mobile application or a web interface can remotely access the architecture and provide a user with an opportunity to monitor live data and control security equipment regardless of where he or she is. The proposed design concentrates on the dependability, low power usage, and safe transmission of data in a bid to eliminate the risk of unauthorized accessibility of the information in the system. The system enhances the overall home security by ensuring it is more secure and it also provides constant vigilance besides ensuring timely intervention in case of any potential threat due to its scalability feature and easy integration with the already existing home infrastructure [2].

The suggested system presents an Internet of Things (IoT)-based system of home automation and a complex smart security system that will be implemented in the next-generation computing environment. The architecture will also affect the devices of the house, sensors, and control units through a centralized IoT platform that allows faultless communication, automation, and intelligent decision- making. The system allows to control the lighting, appliances, climate and access control automatically and monitor the security associated parameters and attributes such as motion detectors, intrusion attempt and environmental hazards. It has incorporated intelligent authentication, secure communication protocols to ensure privacy of information and prevent unnecessary access. In order to meet the demands of safe computing in the new environment, the scheme resorts to automated threat identification, message encryption transmission, and real-time notifications. Computer safety enhances reliability, scalability, and resiliency of the systems to cyber and physical attacks. The suggested model is designed to be adaptable and maximally utilize the resources, and that is why it can serve next-generation smart houses, incorporating automation, security counterintelligence, and powerful computational security into one platform [3].

The Smart Home Security System is developed on the basis of IoT and ESP8266 and is designed to provide a stable and real-time surveillance and access control in living environments. This system uses as the main component of communication the ESP8266 Wi-Fi board, with the help of which it is possible to make the connection between sensors, actuators, and cloud- based services as seamlessly as possible. This is achieved using different sensors which are motion sensors, door sensors and the environmental sensors that watch the home environment at all times. The ESP8266 will send the alert messages to the homeowner over a web interface / mobile app in case of anomalous activity or intrusion. It is also possible to operate local alarm systems (ex: buzzers or indication LEDs) with the system to produce on-site alarms. The IoT connectivity will enable one to view real time data, test to ensure that the system is

displaying its status and control the connected devices whether in the house or any other place where one can access the internet. The proposed structure has low cost, energy efficient and scalable features, which would suit small and large residential setups. The system can enhance the security of the home through wireless communication systems, real-time and automated alerts, thus, making it easy to the people and help them monitor their homes safely even in the absence of guardian presence [4].

Machine Learning-Driven IoT Safety of Women is geared towards the security of the customer in that it is designed to provide an intelligent and constantly up- to-date safety by providing a complex sensory, connectivity, and predictive analysis. The system is connected to wearable or portable IoT hardware where sensors are running such as GPS, accelerator, microphone, and emergency trigger modules to continuously scan the environment and circumstances. Having machine learning algorithms available will allow scanning movement and voice stress patterns or indicators of abnormal behavior to come to the conclusion about potential dangerous and distressing situations. In situations that the system identifies any suspicious activity or in instances when the user activates the system, an emergency notification will be sent to the designated recipients or charge in-charge of security, with a live location data. The IoT model also ensures the fluent nature of the communication between the device and the cloud-based services, and real-time tracking and speedy response are obtained the decisions are also more accurate with the use of machine learning, as the number of false alarms is reduced, and it adjusts itself to the habits of the user. The suggested system was intended to be portable, reliable, and low power consumption, among others, which presents a proactive and intelligent manner of improving the personal safety of the women of both the urban and remote set-ups of the same city [5].

II. RELATED WORKS

The Enhanced Smart Doorbell System Based on Face Recognition is developed to benefit the security of the home by providing intelligent recognition of the visitors and automatic controlled access. The system integrates an advanced camera with face recognition algorithm to check and verify people who stand by the entrance. Images of users who are captured are processed through the machine learning methods in real time to match the facial attributes with a pre-registered database of good users. The system can potentially unlock or alert the homeowner under a mobile application on a face recognition. The instance of some stranger or unknown person will provide immediate signals and shots taken to the user to confirm the case. The system can also be competent to make alarm system such as buzzers or even visual cue in order to enhance on site awareness. As a device that is to be integrated with the IoT, the smart doorbell is known to offer the user an opportunity to monitor the object remotely, provide video streaming in real-time, and transfer data safely. The enhanced version should target the higher recognition accuracy, reduced false identifications, and the effective processing which can be embedded into the embedded systems. The system provides a safe and modern approach to controlling access to the homes with smart facial recognition to offer a safe and reliable access into the home along with automatic communications and notification systems [6].

The Doorbell Security System of Doorstep Crime Prevention is a security system that helps to improve the comfort of people in their homes by keeping an eye on the people that enter the home by verifying them. It comprises a camera, motion detection sensor, and real time communication technology which are combined together to ensure that the door step area is constantly monitored. Once a visitor is detected, the system takes pictures or records videos and sends the data to the homeowner via a private mobile or web application. The suggested framework is interested in recognizing suspicious activity and unauthorized access attempts prior to the physical touch. Much more sophisticated capabilities like facial recognition, two-way audio video conferencing, and automatic alert messages allow remote verification of visitors and proper response by homeowners. The system may as well trigger local alarms or notify intended contacts in situations involving emergencies. Through surveillance, remote access, and intelligent monitoring, the system can serve as a proactive measure to doorstep crimes, like theft or impersonation or forced entry. It has an IoT-based and scalable structure to guarantee 24/7 monitoring, data security and stability, and works well in the modern home security management [7].

The Door Hinge Crime Prevention Doorbell Security System is a security system that assists in enhancing the comfort of individuals in their home by monitoring the individuals that get into the house by checking them. It has a camera, motion detector sensor and the real time communication technology that are integrated together to make sure that the door step area is continuously monitored. After a guest has been identified, the system captures images or documents video and transmits the information to the owner of the home through a personal mobile or web-based application. The proposed model is also concerned with the identification of suspicious activity and unauthorized access attempts before the physical contact. A lot more advanced features such as face recognition and two-way video audio conferencing as well as automatic alerts enable remote checking of visitors and adequate response by homeowners. The system can also activate local alarms or send relevant contacts in cases of emergencies. The system can be used as a preventive action, however, to doorstep crimes, such as theft or impersonation or forced entry through surveillance, remote access, and intelligent monitoring. It possesses an IoT enabled and scalable architecture to ensure round the clock monitoring, data integrity and reliability, and is compatible in the new home security management [7].

The proposed system will integrate the home automation of the IoT with a smart security system to adopt secure and smart next-generation environments. It may be exploited to offer centralized management of the domestic appliances, environment management, and real-time surveillance of the security parameters. This system has an advantage of remote access over mobile platform or through web platform and also has advantage of encryption and secure authentication of communication. A combination of safe computing and constant monitoring ideas and automation increases efficiency, security of the data, and overall reliability of the existing smart homes based on the framework. [8].

The suggested IoT Smart Home Security System and Face Recognition is aimed at enhancing the safety of the residents and raising the degree of smartness control and automated identity verification. The system will integrate iot enabled devices such as cameras, sensors and microcontrollers to

provide the use of round-the-clock surveillance of the points of entry and the surrounding environment. When a visitor approaches the camera, the camera takes a record of the face images and they are founded on the principle of face recognition algorithms to find the match of an entity that is stored in a database as an authorized individual. The system could either be automatically activated to grant the access or alert the homeowner in case the face scanned belongs to a registered user. The user receives instant notification and taken images where there are unknown or suspicious individuals both via a mobile or a web platform. This enables checking of the distant location and action in case of threats in time. The IoT system is able to ensure real- time delivery of information, remote connection and data security. The devices and smart face recognition that are attached serve the goal of reducing the number of unauthorized entries, maximizing the effectiveness of the control systems, and providing a flexible and dependable approach to the issue of the smart home security today [9].

Design and Development of an Intelligent Home with Automated Environmental Control has a purpose of designing a smart house system which is capable of monitoring the conditions in the home automatically and automatically regulating it. The specified scheme integrates the sensors and microcontrollers and communication blocks to measure the data pertaining to the temperature, the humidity, the air quality, and the light levels to provide full-time monitoring of the aspects. Based on the real-time data analysis, the system will automatically control similar equipment as fans, air conditioners, heaters, ventilations systems and light units that will be necessary to secure the optimum degrees of comfort. The mechanism of control is based on the predefined threshold or adaptation algorithms, which dynamically respond to the environmental changes. The intelligent home model can also have remote monitoring, can include a manual override with the assistance of the mobile or web-based interface which ensures flexibility and ease of use of the item. They are concerned with the saving in regards to the energy use, reduction of the human intervention and regularity of the system operation. The system has provided a long-term and convenient resolution to the modern residential environmental management by incorporating automation, sensor- based, monitoring and intelligent control devices [10

Voice Recognition System of a Smart Doorbell aims at enhancing the security of the homes and ease of use by the individual using speech recognition and dialogue. The system will be made up of microphone, embedded processing unit and wireless communication interface to record and process voice inputs of visitors in the entrance. The system then matches speech patterns of the individual with a registered database of known users based on voice recognition algorithms. The doorbell system will be able to notify the homeowner or give access when a match is detected. The system will also alert the user when the voice is not recognized besides giving the audio samples that will be recorded. It will also enable two-way communication, where the homeowners would be in a position to remotely communicate with the visitors through a connected mobile or web application. It involves emphasis on proper speech processing, noisy data processing algorithms, safe data transfer and real-time response. The suggested system may provide a safe, touchless, and intelligent solution to the modern smart home access management by integrating voice-based authentication and IoT connectivity [11].

The Smart Door Access System and the Fingerprint Biometric System are designed, in order to provide the secure and reliable access control, according to the individual physiological recognition. The system incorporates fingerprint sensor and processing unit through the utilization of microcontrollers whereby fingerprint pattern of the user is captured, processed, and authenticated to use the system. In its usage, the sensor scans the fingerprint of a user and converts them into a digital template, which is compared to the biometric data, already stored in the system database. To open the door, the control unit will turn on a motorized locking mechanism in case a valid match is found. The system can send alert messages where there is unauthorized access, the system can switch on a buzzer or the system can record the incident that will be monitored. The suggested framework concentrates on the correct high authentication, low chances of duplication and low amounts of manual interaction. This system offers a better degree of access control with the integration of the biometric verification and embedded control with the optional IoT access to be able to monitor the user remotely without making this particular system too inconvenient. It is a dependable, and scalable residential, institutional and commercial door security use [12].

The proposed New Smart Home Intruder Detection System Based on Deep Learning is developed to offer increased security to a home through smart surveillance and automatic threat detection. The system will be configured to unite cameras and motion sensors with a processing unit that can be trained on the concept of deep learning to analyze the visual information in real-time. The system detects the normal household motions and the suspicious or unauthorized motions with the assistance of the advanced neural network models. Various data sets are used to optimize the deep learning model to improve the detection rate and the rate of the false alarm that is due to pets, environment, or habitual movement. Upon anticipation of an intrusion, the system will automatically generate the alert through the sending of mobile notifications, alarms, or cloud-based application so that the user would respond. To make it reliable and confidential, the proposed structure allows uninterrupted control, access at any point, and secure transfer of data. The system will provide an adaptive, scalable, and high-precision intrusion detection that will apply to a modern smart home environment through the introduction of a combination of the IoT connectivity and deep learning approaches [13].

The suggested Home Security System with Face Recognition via Convolutional Neural Network (CNN) is a system, which is targeted to provide intelligent and automated access control to the home environment. This system is a combination of a camera sub-unit and CNN-based facial recognition model to identify persons entering the door at the entrance in real-time. The acquired images are fed through many convolutional layers in order to detect the distinguishing characteristics in the faces and compared to a stored database of verified users. The CNN model is superior in the learning of the recognition accuracy in the complex facial patterns under various conditions of light, angles and expression. On identification of a familiar face (registered face), the system can automatically open the door, or can send an alert to the homeowner. Real- time notifications and images captured are remotely transferred in the case of the identification of an unknown individual through an associated IoT platform to be examined. The framework is aimed at safe information processing, real-time processing, and reduced false recognition. The solution is

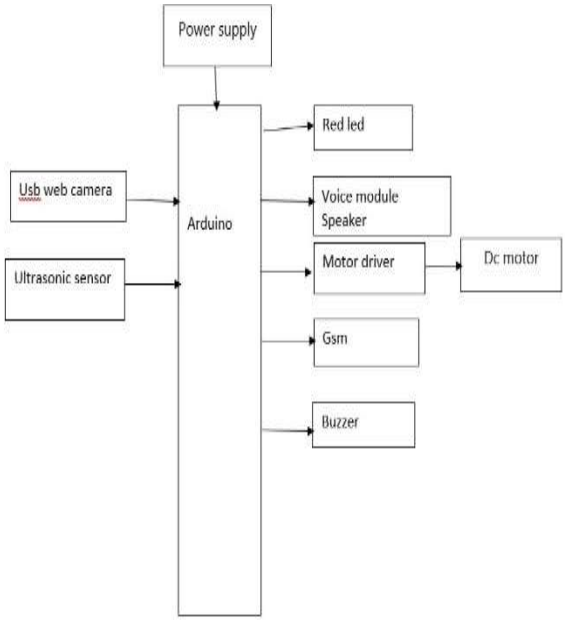
provided to the current home security control, where it is reliable, scalable, and efficient because it includes deep learning-based facial recognition, as well as IoT- related surveillance [14].

The provided system is oriented to the improvement of the surveillance within smart homes and implies the usage of the YOLO (You Only Look Once) object detection model that is extended in regards to the transfer learning and model quantization approaches. It is a framework of a combination of surveillance cameras and embedded processing unit capable of detecting objects and identifying threats in real-time. Using the transfer learning, the YOLO model can be refined on the domain-specific datasets, which enables the model to detect the objects of interest more appropriately such as the intruders, suspicious activity, or restricted objects within the households. The size of the model and processing complexity is minimized using quantization without significantly affecting the accuracy of detection in order to maximize the amount of computational performance. This enables its efficient implementation on resource constrained edge devices commonly found in smart home implementation. The integrity of the detection speed optimization, and the increased accuracy of classification can ensure continuous monitoring with a minor latency. The system supports some of the features such as real-time alert, remote access through the IoT platform, and efficient communication of data. Integrating the deep learning optimization algorithms with other ones, the proposed system will improve the accuracy and effectiveness rates, which justifies a scalable and intelligent smart home surveillance system of the new generations [15].

III. PROPOSED METHOD

The Smart Doorbell System proposed improves the security and control of the user by incorporating an Arduino that has an embedded Python to provide intelligent automation. Upon the arrival of a visitor, an ultrasonic sensor is used to identify his/her presence which activates the USB camera to take a photograph and transfer it to the homeowner using Telegram to validate it in real-time. The system with a single approval message enables a voice module that shows whether it will be really approved, and opens the door automatically and through a motor driver which opens the door after a specified time. Besides the remote access control, it has a GSM module, a buzzer, and a red LED that give the necessary signals when the access is unauthorized, or in case of emergency situations. The system operates on a permanent 12V supply and is linked together with high quality connectors and forms an automated reliable and customizable home entry system that outsmarts the shortcomings of the current systems.

IV. Block Diagram:



V. Methodology

A. Principle of Functioning

The Smart Doorbell System is predetermined by several postulates of sensor-based recognition of a visitor and verification of the picture and the detection of a user in real time. The system will consist of integrated Python and Arduino hardware management to enable smart monitoring, safe communication and automated door movement. The ultrasonic sensor is not limited and monitoring of the entry is done and the visitor presence is recognized within an established range of distances. On receiving a signal, a USB web camera begins to take a picture. The obtained photograph is transmitted through the internet-based communication platform (Telegram) to the house owner to be identified remotely. As soon as the homeowner has gone through the image he/she can authorize through an authorization message. Upon the confirmation being received, the embedded controller activates the voice module to inform and instruct the motor driver to release the door and open it. The door will be left open at a predetermined time that is 20 seconds after which it automatically closes to create some form of security. To enhance the level of security, the system is further equipped with the addition of GSM to send alert messages upon an effort to intrude in accessing it illegally or an emergency. It also has a red LED and a buzzer that provides a quick visual and audible notification to alert about a suspicious activity. The entire system is operated on a regulated 12 V direct current and appropriate control of voltage to achieve stable, continuous and consistent operation. This type of sensing, communication, authentication and automated actuation combination solutions will provide a secure, interactive, and efficient entry management solution to a smart home.

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B. Hardware & Alerts:

The Smart Doorbell System is built with an Arduino based sensor management unit which incorporates Python processing to give it intelligent control and communication. This is done by the presence of an ultrasonic sensor that continuously scans the area in front of the door, therefore indicating the presence of a visitor. The USB web camera will take a live image of the visitor and transmit it through the internet to the homestead, using Telegram, to confirm the identities of the homeowner, remotely. When the owner of the house examines the photo obtained, one can give access by replying to it with a confirmation message via Telegram. Once the authorization has been received, the embedded Python controller transmits a message to the Arduino to provoke the voice module to inform and to send an Arduino a signal to unlock and open the door with the help of the motor driver. The door remains open within a given duration of 20 seconds after which the door then closes automatically as a security measure. Just to be at the safe side, the device is provided with a GSM module that will be used to send an alert message in case an emergency event or any suspect activity occurs. A buzzer and a red LED indicate the presence of an audible and visual notification on accessing unauthorized areas. The coordinated operation of sensors, camera inspection, user recognition and the controlled door will suggest the secure and interactive smart entry control system which is suitable to the modern residential application.

C. Power Requirements:

The whole system will be powered by a controlled 12 V DC power adapter to provide a stable and continuous operation. The Arduino board, ultrasonic sensor, GSM module, motor driver, buzzer, LED indicators and communication modules are supplied with the help of proper voltage regulation circuits. The motor driver is supplied with adequate current that can

be used to open and close the door locking and unlocking mechanism. The correct power management guarantees the stable real-time monitoring, safe communication, and fluid door control whilst being energy efficient to implement a smart home constantly.

D. Performance Comparison Table:

Parameter	Specification / Metric	Description
Central Controller	Arduino Uno with Embedded Python Integration	Acts as the primary control unit that processes ultrasonic sensor data, communicates with the Python-based system, manages Telegram communication, and controls door operation mechanisms
Visitor Detection Sensor	Ultrasonic Sensor	Continuously measures distance to detect the presence of a visitor near the door and triggers the image capture process.
Image Acquisition	USB Web Camera	Captures real-time images of visitors and sends them to the embedded Python system for remote transmission via Telegram.
Communication Platform	Telegram Bot Integration	Enables real-time image transmission to the homeowner and receives authorization commands for remote door access control.
Door Actuation System	Motor Driver with DC Motor	Controls the locking and unlocking mechanism of the door based on user authorization, ensuring smooth and secure door movement.
Voice Notification Module	APR Voice Module / Audio Module	Provides audible feedback or welcome messages when access is granted or system events occur.

Alert System	Buzzer and Red LED	Generates audible and visual alerts during unauthorized access attempts or emergency situations.
Remote Alert Communication	GSM Module	Sends SMS notifications to the registered mobile number during suspicious activity or system emergencies.

Table 1 *Performance Comparison Table*

Smart Doorbell System is being based on the platform of Arduino Uno investigation with embedded Python program to ensure the functionality of hardware control and intelligent communication. The Arduino is the processor that controls the sensor input, hardware control and coordination to the remote user interface. The concept of visitor detection is achieved through the ultrasonic sensor that has been continuously used to measure how far the people are to the door. When the range detected is within a set range, the system detects the presence of a visitor and begins further processing. Real time picture of the visitor is then taken using a USB web camera which is sent to embedded Python module where it is sent via internet. In long distance communication, Telegram bot is used in the application. The obtained photo is sent at once to the Telegram account of the homeowner and, therefore, can be checked on the fly. Depending on the image received, the user can send a command of authorization. When a confirmation is obtained, the Arduino drives the motor driver that is connected to a DC motor to unlock the door by driving the motor. The system involves an automatic process of the timing that leaves the door open at 20 seconds automaticity closes the door, hence keeps the security. The design will include voice module so that the interaction becomes user friendly and the user will receive audible notification on different occasions as the door is being used or a visitor is granted access. Security is also boosted using the help of buzzer and red LED indicators that give instant warning whenever one wants to access the object without authorization. The GSM module will also be included which will permit sending of SMS to the mobile number that has been registered to provide notification in case of any suspicious thing or in case of emergency cases. A controlled supply (12 V DC) has been adopted to electronic components are working in a similar manner. The sensors, communication modules and motor driver are fed by the voltage regulation circuits with the correct amount of power to ensure the consistent and reliable control of smart-house entry.

E. Reaction time of the system components

Table 1: Comparative Analysis of Conventional Techniques and the Developed Approach

Parameter	Existing Methods	Proposed Approach (Our System)
Visitor Detection	Conventional doorbells or simple motion sensors without remote verification	Ultrasonic sensor detects visitor presence and triggers real-time image capture for remote verification

Monitoring Method	Local or in-person monitoring only	Continuous remote monitoring through Telegram for real-time visitor alerts and access control
Access Authorization	Manual key or physical permission	Remote user verification via image review and Telegram-based authorization messages
Alert Mechanism	Simple doorbell chime	Multi-level alerts including buzzer, red LED, and voice notification for visitor presence or unauthorized access
Door Operation	Manual unlocking and opening	Automated door unlocking and opening via motor driver controlled by Arduino upon remote user confirmation
Unauthorized Access Handling	Limited to local alarms or physical intervention	GSM module sends SMS notifications during unauthorized access attempts, combined with buzzer and LED alerts

Table 2: Smart Doorbell System is being based on the platform of Arduino Uno investigation with embedded Python program to ensure the functionality of hardware control and intelligent communication. The Arduino is the processor that controls the sensor input, hardware control and coordination to the remote user interface. The concept of visitor detection is achieved through the ultrasonic sensor that has been continuously used to measure how far the people are to the door. When the range detected is within a set range, the system detects the presence of a visitor and begins further processing. Real time picture of the visitor is then taken using a USB web camera which is sent to embedded Python module where it is sent via internet. In long distance communication, Telegram bot is used in the application. The obtained photo is sent at once to the Telegram account of the homeowner and, therefore, can be checked on the fly. Depending on the image received, the user can send a command of authorization. When a confirmation is obtained, the Arduino drives the motor driver that is connected to a DC motor to unlock the door by driving the motor. The system involves an automatic process of the timing that leaves the door open at 20 seconds automaticity close the door, hence keeps the security. The design will include voice module so that the interaction becomes user friendly and the user will receive audible notification on different occasions as the door is being used or a visitor is granted access. Security is also boosted using the help of buzzer and red LED indicators that give instant warning whenever one wants to access the object without authorization. The GSM module will also be included which will permit sending of SMS to the mobile number that has been registered to provide notification in case of any suspicious thing or in case of emergency cases. A controlled supply (12 V DC) has been adopted to supply

power to the whole system to ensure that all the electronic components are working in a similar manner. The sensors, communication modules and motor driver are fed by the voltage regulation circuits with the correct amount of power to ensure the consistent and reliable control of smart-house entry.

VI. RESULTS

FIG1-

Fig1 – *Hardware Assembly*

FIG2 –

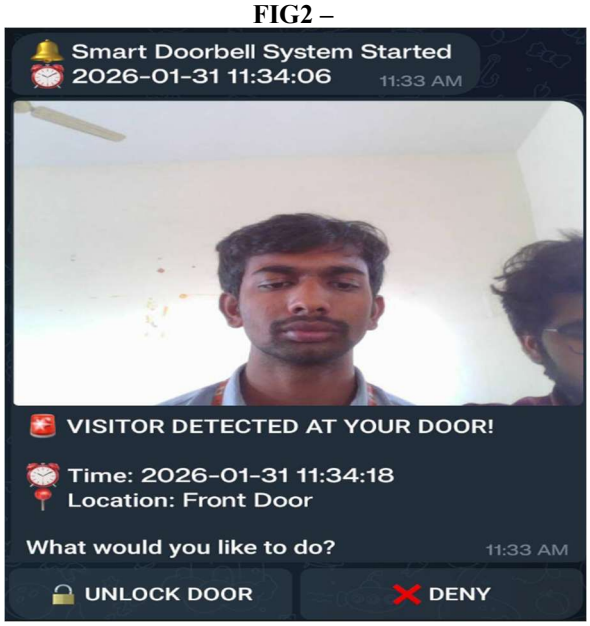

Fig2 – Software Working

VII. CONCLUSION

The Smart Doorbell System presented in this project provides an effective and secure solution for modern home access management. By integrating Arduino- based sensors with embedded Python control, the system enables real-time visitor detection, image capture, and remote verification through Telegram. User-authorized access ensures enhanced security and reduces the risk of unauthorized entry. The automated door operation, along with safety features such as GSM alerts, buzzer, and LED indicators, improves system reliability and responsiveness during emergency situations. Overall, the proposed system offers a user-friendly, interactive, and cost-effective approach to home security, demonstrating the potential of IoT-enabled and embedded systems in smart home applications.

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